OOAD PROJECT- AGRICONNECT

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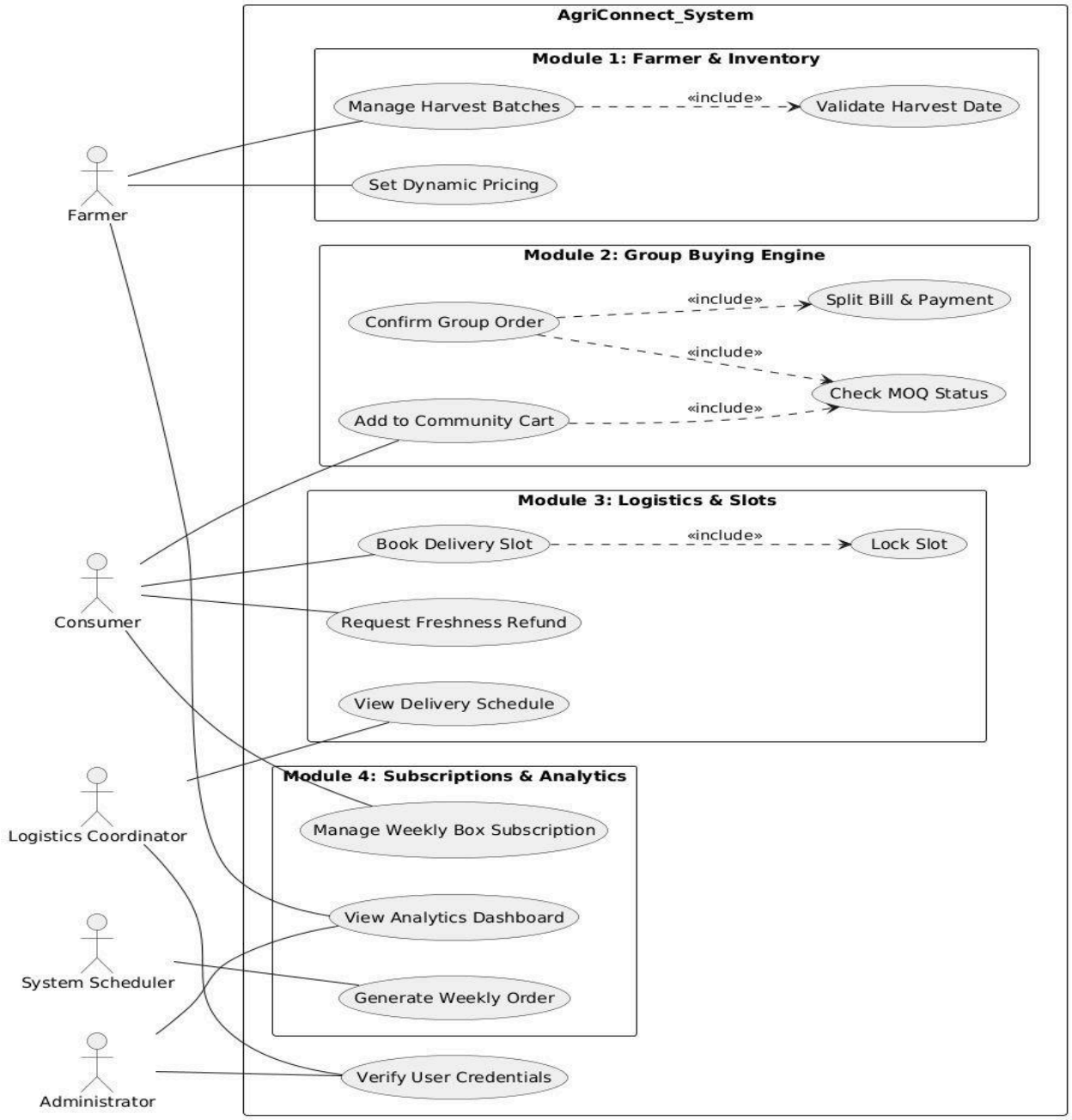
USE CASE DIAGRAM:



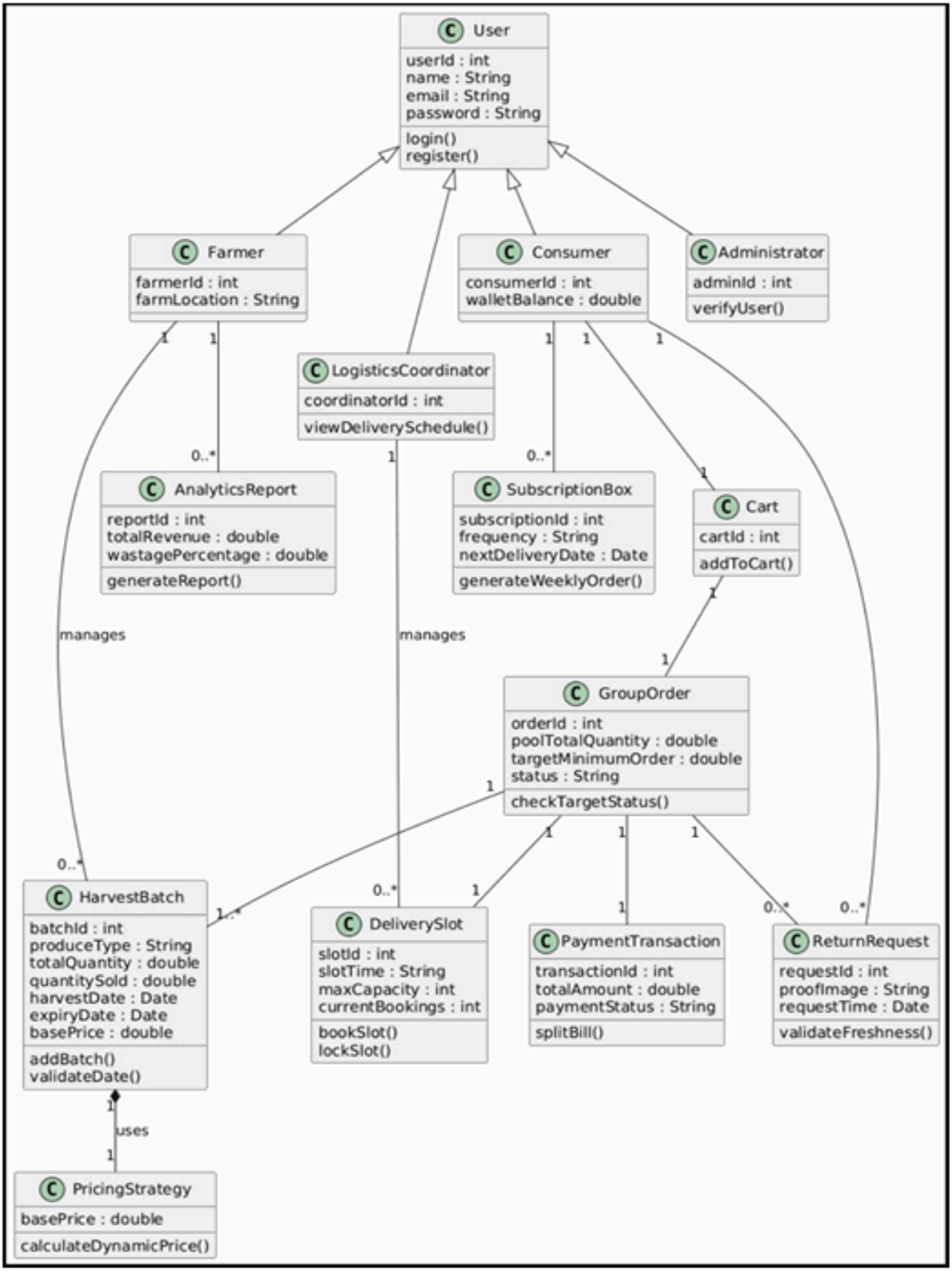
Farmer: Manages the supply side, including harvest batches and dynamic pricing.

Consumer: Handles the demand side, such as community carts, delivery slots, and refund requests.
Logistics Coordinator & System Scheduler: Manage background operations like delivery schedules and automated weekly orders.
Administrator: Ensures system integrity via user verification.
Key Relationships: Note the <<include>> relationships, such as "Manage Harvest Batches" requiring "Validate Harvest Date," which ensures data integrity at the entry point.

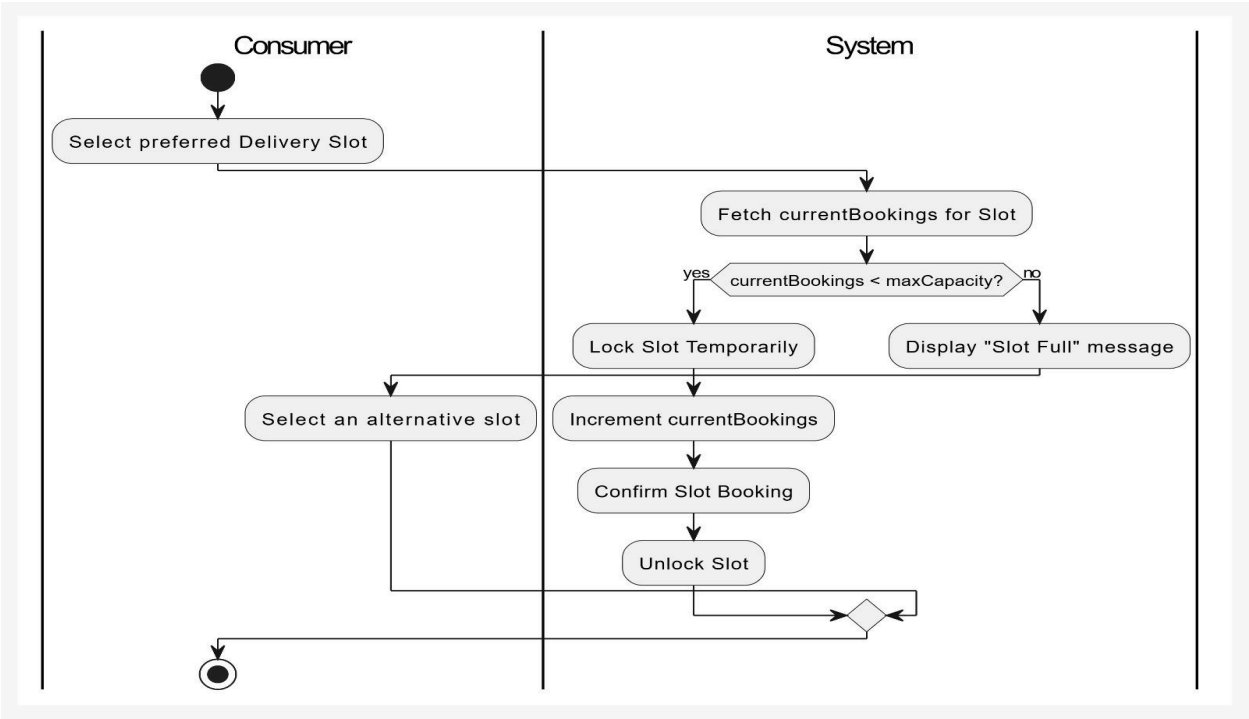
The Class Diagram represents the **static structure** of the system, showing the attributes and methods of your objects.

- **Inheritance:** We have a User base class with specialized children (Farmer, Consumer, LogisticsCoordinator, Administrator), which is a perfect application of the **Open-Closed Principle**.
- **Composition & Association:**
    - **HarvestBatch & PricingStrategy:** A batch "uses" a strategy to calculate prices dynamically.
    - **GroupOrder:** This is the heart of your "Module 2," managing the poolTotalQuantity and targetMinimumOrder (MOQ) logic.
- **Multiplicity:** For example, one Consumer can have multiple SubscriptionBox entries (0..*), showing a one-to-many relationship.

Class Modeling



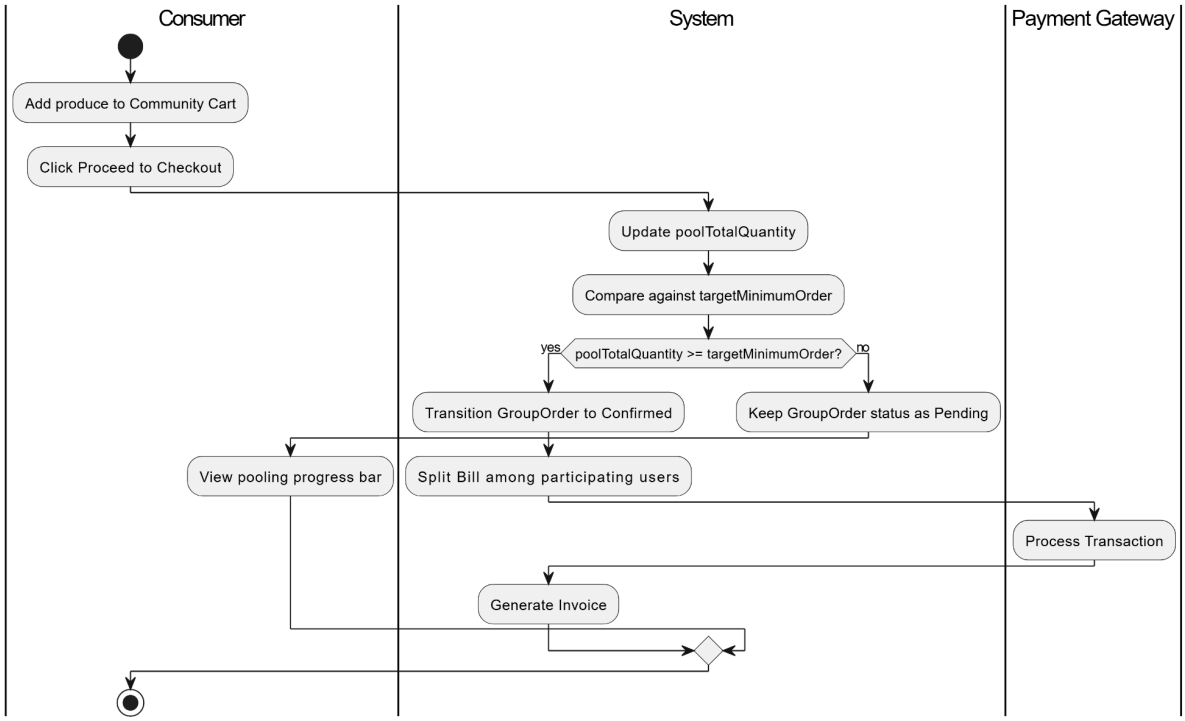
ACTIVITY DIAGRAM: (KHUSHI MAHESH)



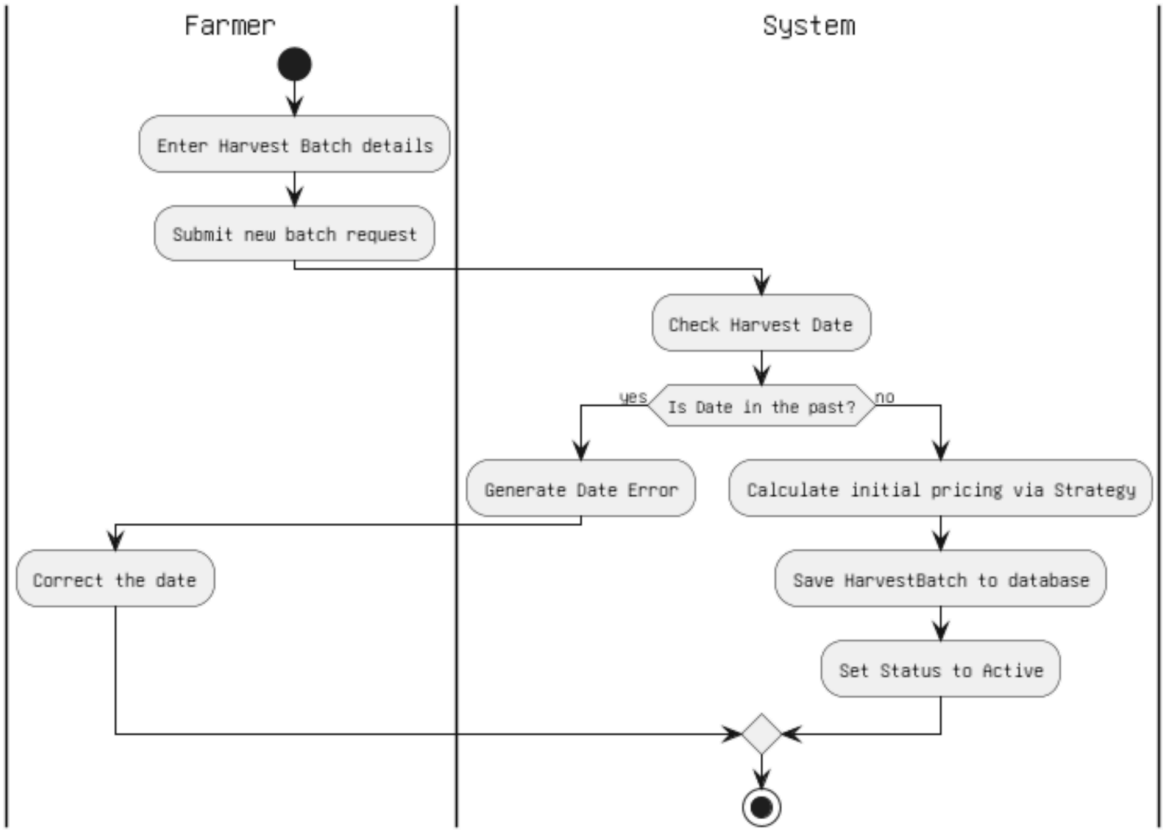
These represent the **workflow/business logic**. Currently, your PDF shows 2 distinct workflows (Delivery Slots and Subscription Processing).

- **Delivery Slot Booking:** Shows the logic for checking currentBookings against maxCapacity. It includes a decision node to "Lock Slot Temporarily" or display a "Slot Full" message.
- **Subscription Scheduler:** A background process that "wakes up at midnight" to process recurring orders, check wallet balances, and handle insufficient funds.
- **Grading Tip:** The guidelines require **4 Activity Diagrams**. You should add two more, such as "Add to Community Cart/Group Buying" and "Farmer Batch Upload & Validation."

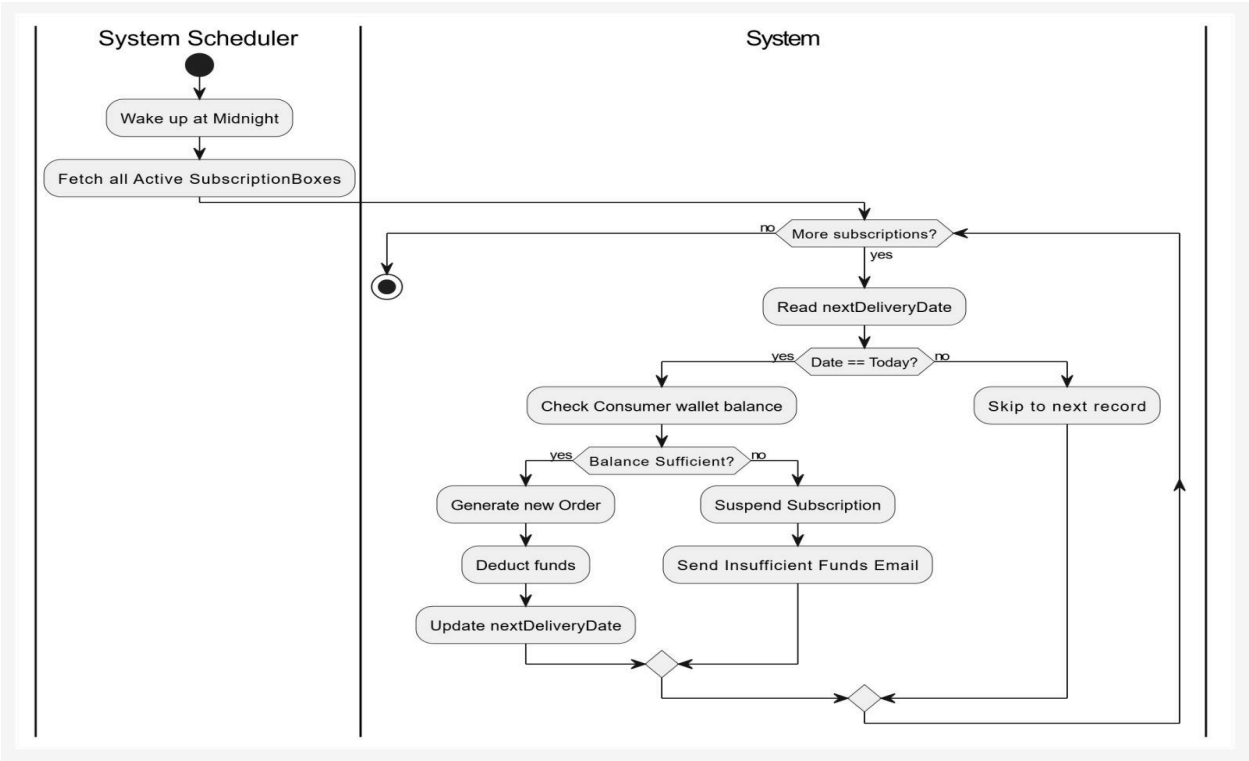
(KHUSHI BHUPESH)



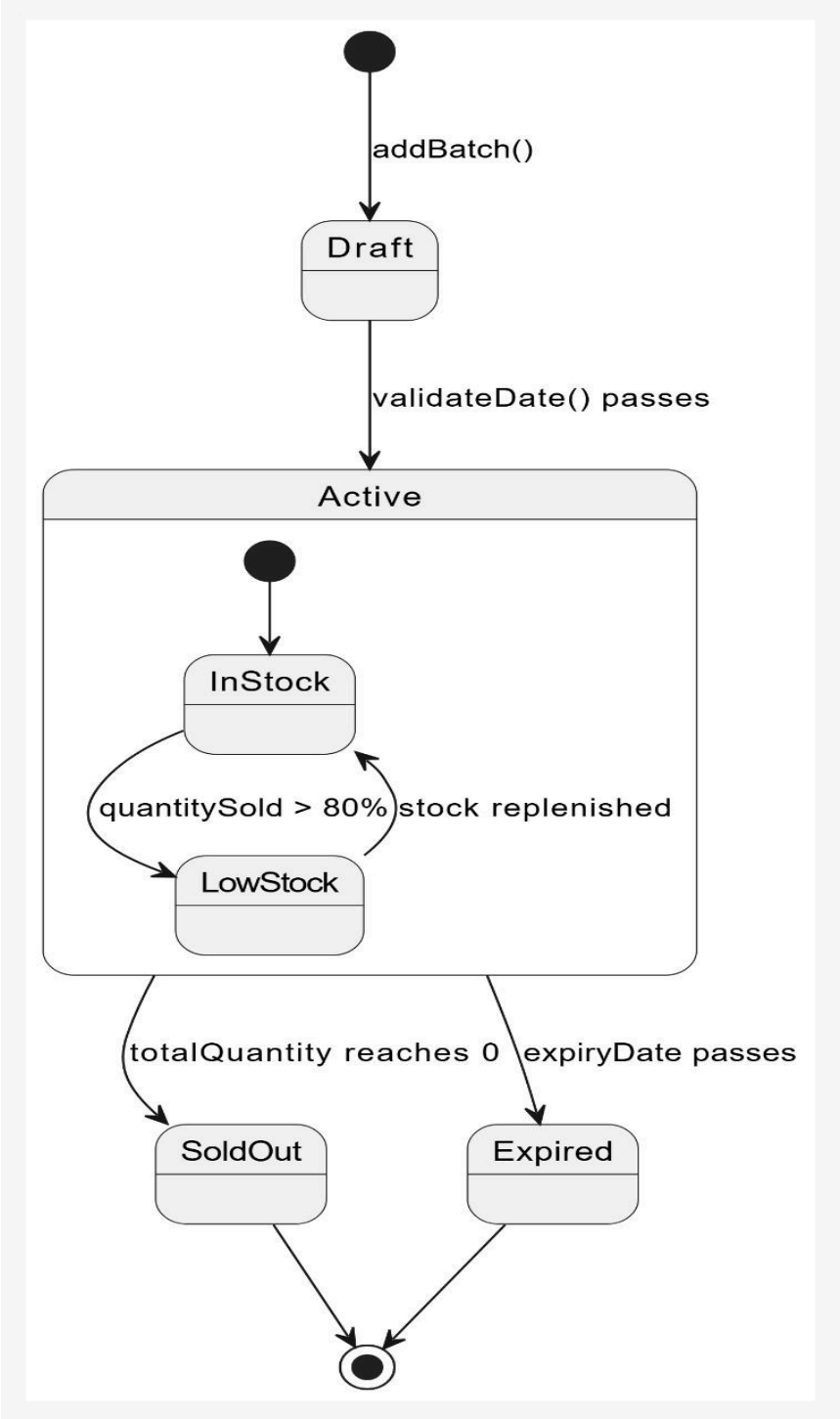
(KARTIK SHARMA)



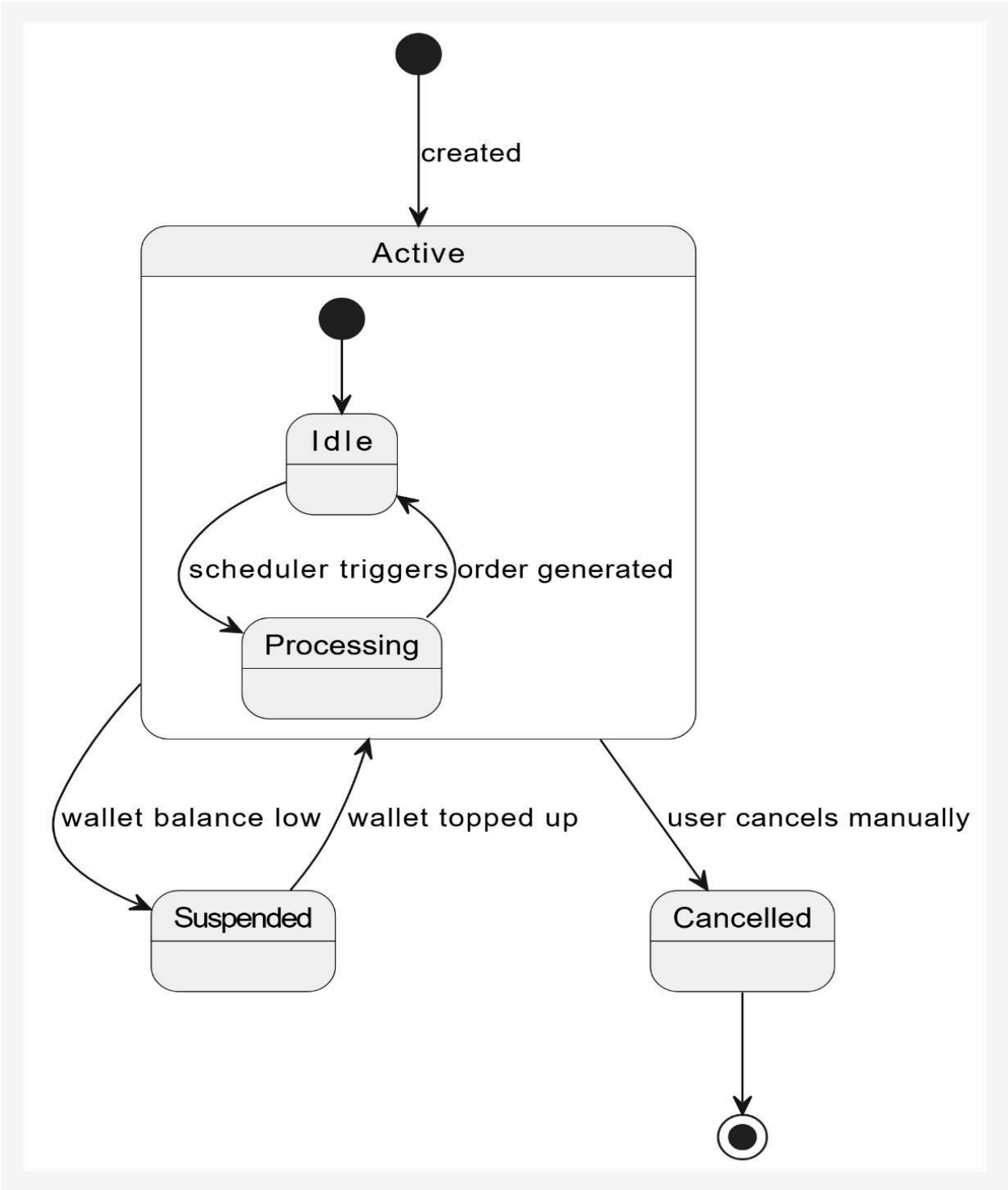
(KAVERI SHARMA)



STATE DIAGRAM:
(KARTIK SHARMA)



(KAVERI SHARMA)



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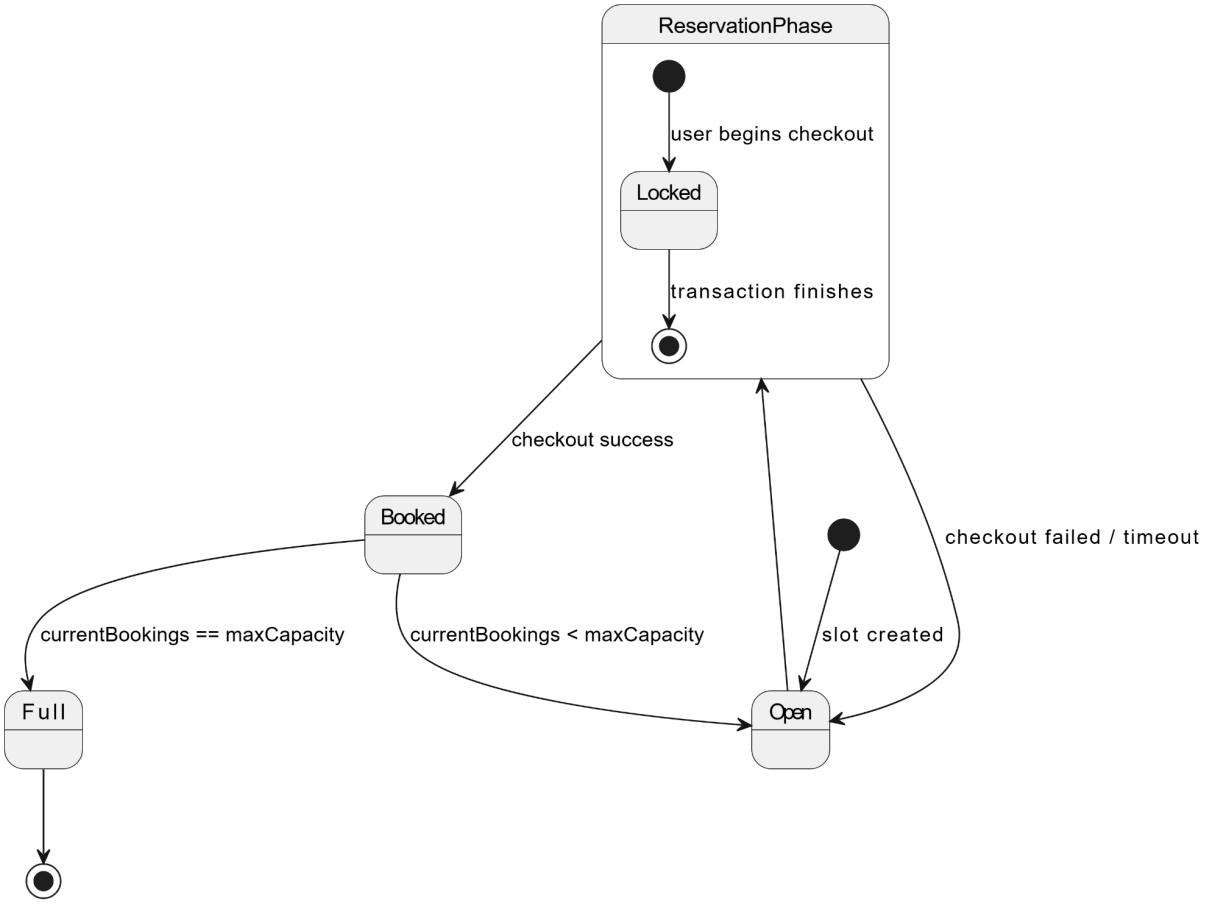

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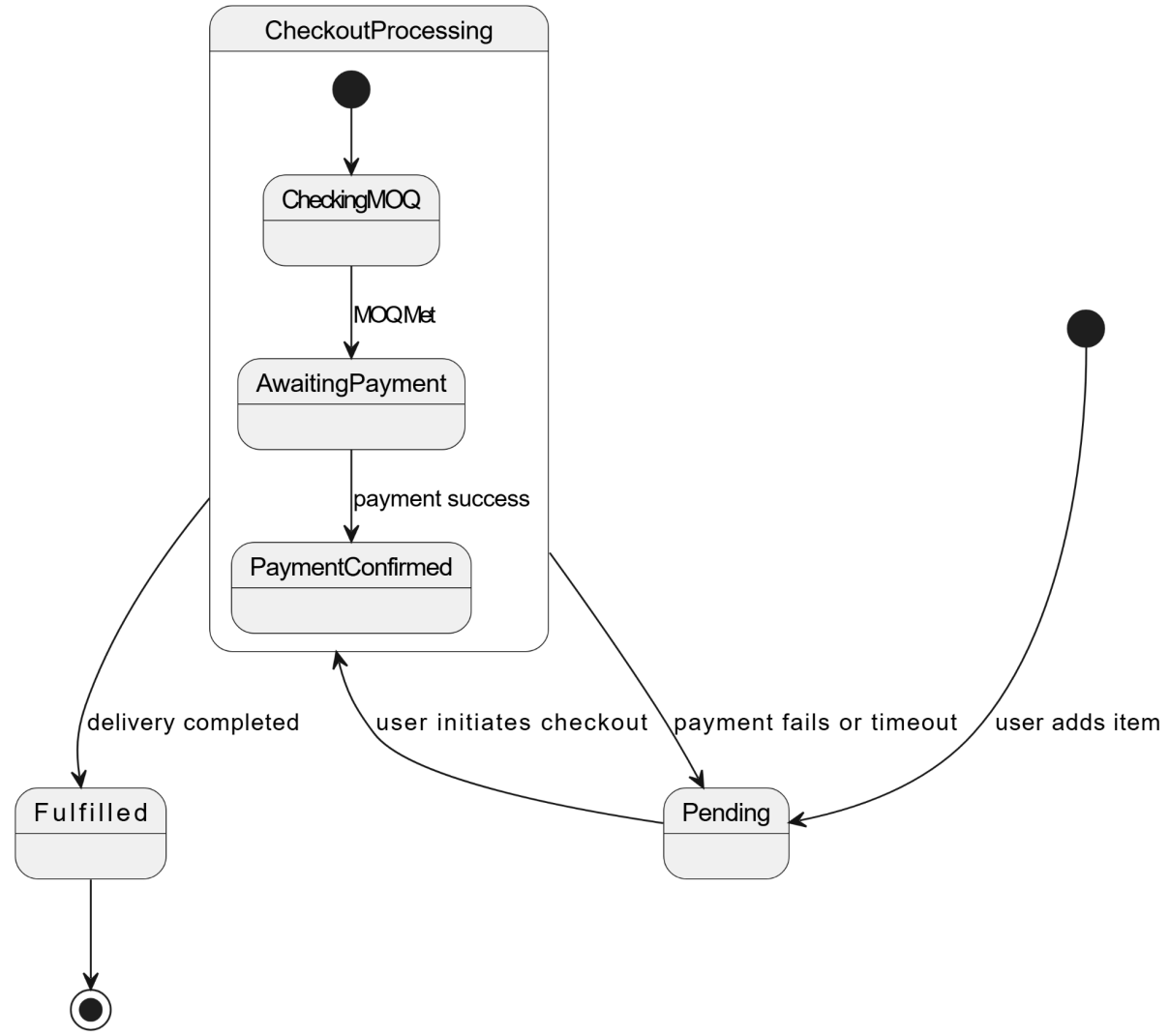

These track the **lifecycle** of a single object.

- **HarvestBatch State:** Transitions from Draft $\rightarrow$ Active (InStock/LowStock)-> SoldOut or Expired.
- **Checkout/Order State:** Tracks the progress from Checking MOQ to PaymentConfirmed and finally Fulfilled.
- **Subscription State:** Shows the Active-> Suspended (if balance is low) -> Cancelled flow.
- **Grading Tip:** You need **4 State Diagrams** for full marks. Consider adding a state diagram for "Refund Requests."